
Future Interns.

Task 3 - Create and optimize AI prompt for content generation .

Prompt Engineering – Comparative Analysis

April, 2025

➤ Objective

To begin this project, I first decided to develop three simple prompts. The second step is to make them as detailed, clear, concise and practical as possible in order to achieve better results. Finally, I will write the expected answer and later compare it to the AI's actual response. To enhance precision, two different AIs - **Claude 3.5 Sonnet** and **GPT 4o mini** - were used and both results will be written down below.

1st prompt

Lately I've been interested in machine learning and would like to understand the basics. Tell me what it is and where to begin. Do all of it while pretending to be an engineer giving a quick intro in a workshop.

2nd prompt

A friend of mine plans to open a cat cafe but is struggling in finding names. Come up with a few creative suggestions having in mind that her name is Lisa, and she owns two kitties called Hades and Athena.

3rd prompt

I'm struggling to write an ad copy for a small graphic design business specialized in company logos. Please create a compelling and easy-to-understand copy that appeals to established businesses looking for an ideal brand identity.

➤ Expected answer vs actual response

1st prompt

Expected: a youthful and simple language covering the basics of ML. Should avoid too much technical terms without leaving out crucial concepts.

Responses (Claude's and GPT's, respectively):



Adjusts imaginary glasses and clears throat

"Hey everyone, thanks for joining this intro workshop on Machine Learning! I'm excited to break this down for you in the next few minutes.

Walks to imaginary whiteboard

So, at its core, Machine Learning (ML) is a subset of artificial intelligence that enables computers to learn from data without being explicitly programmed. Think of it like teaching a child - instead of giving them strict rules for everything, you show them examples, and they learn to recognize patterns.

Let me give you a quick example: Imagine teaching a computer to distinguish between cats and dogs. Instead of writing rules like 'if it has pointy ears and whiskers, it's a cat,' we show the computer thousands of labeled pictures of cats and dogs. The computer then learns to identify the patterns that make a cat a cat and a dog a dog.

Picks up marker

For beginners, I recommend this learning path:

1. Prerequisites:
 - Python programming (start with courses on Codecademy or Python.org)
 - Basic statistics and linear algebra
 - Data analysis basics
2. First Steps:
 - Take Andrew Ng's Machine Learning course on Coursera
 - Practice with scikit-learn library
 - Start with simple projects like:
 - Linear regression
 - Basic classification problems

Turns back to audience

Pro tip: Don't get overwhelmed by the math initially. Focus on understanding the concepts and gradually dive deeper.

Any questions before we move to the hands-on part of the workshop?"

Looks expectantly at the audience

But where's the role-playing?



Apologies for that! Let's try this again with a more engaging, role-playing approach.

[Scene: A workshop room filled with eager participants. The engineer stands at the front, ready to introduce the topic.]

Engineer: "Hello everyone! Welcome to our workshop on machine learning! I'm thrilled to see so many of you here, ready to dive into this fascinating field. My name is Alex, and I've been working as a machine learning engineer for several years now. Today, I'm going to give you a quick overview of what machine learning is and how you can get started on your journey.

[Pauses for a moment, looking around the room.]

Engineer: "So, let's start with the basics. Imagine you have a computer that can learn from data, just like we do. That's essentially what machine learning is! Instead of writing specific instructions for the computer to follow, we provide it with data, and it learns to make decisions based on that data.

[Gestures to a whiteboard with a simple diagram of data flowing into a model.]

Engineer: "Here's a simple way to think about it: We feed the machine data—like images, text, or numbers—and it uses algorithms to find patterns. For example, if we want to teach a model to recognize cats in photos, we show it thousands of pictures of cats and non-cats. Over time, it learns to identify the features that distinguish a cat from other objects.

[Nods enthusiastically.]

Engineer: "Now, let's talk about some key concepts. First up, we have data. It's the lifeblood of machine learning. The more quality data you have, the better your model will perform.

[Points to a chart showing different types of data.]

Engineer: "Next, we have algorithms. These are the mathematical models that process our data. Think of them as the recipes that guide the machine in learning. Some popular algorithms include linear regression, decision trees, and neural networks.

[Raises a finger for emphasis.]

Engineer: "And then there's the process of training and testing. We split our data into two parts: one for training the model and the other for testing how well it learned. It's like studying for an exam and then taking the test!

[Pauses to let that sink in.]

Engineer: "Now, let's not forget about overfitting and underfitting. Overfitting is when our model learns the training data too well, including the noise, and fails to generalize to new data. Underfitting is when it's too simple to capture the trends. It's all about finding that sweet spot!

[Looks around the room, engaging with the audience.]

Engineer: "So, where do you start? First, I recommend learning some programming basics, especially in Python. It's the go-to language for machine learning. Then, get comfortable with libraries like Scikit-learn for traditional ML and TensorFlow or PyTorch for deep learning.

[Gestures to a list of resources on a slide.]

Engineer: "Online courses are a fantastic way to learn. I highly recommend Andrew Ng's Machine Learning course on Coursera. It's a great introduction! And don't forget to work on hands-on projects. Platforms like Kaggle are perfect for that.

[Smiles warmly.]

Engineer: "Finally, join a community! Engage with others who share your interest. Whether

Notes: **Claude's** response stood out as the best; complete, creative and immediately understood the role-playing command, delivering the expected result. **GPT's** output, however, missed the engineer persona and delivered a simple introduction to ML. That being said, only after a reminder did **GPT** align with the expectations.

2nd prompt

Expected: here, the intended result was based on creative suggestions of names. Although the cats' names are inspired by Greek mythology, this detail was omitted on purpose to analyze whether the AI would note it independently.

Responses:

 Let me brainstorm some creative cat café names that play with these elements:

Mythological Theme (based on the cats' names):

- Mount Olympus Café
- The Purrslan Oracle
- Nine Lives of Olympus
- The Godly Whiskers

Lisa + Greek Theme:

- Lisa's Pantheon of Purrs
- Lisa's Mythical Meows
- The Grecian Whisker by Lisa

Cat Puns + Mythology:

- Pawlympus
- Catherian Coffee
- The Purrropolis
- Meowthology

Sophisticated/Elegant:

- Café de Les Chats Divins (The Divine Cats)
- The Celestial Feline
- Lisa's Divine Companions

Playful:

- Hades & Athena's Hideout
- The Wise Cat & The Underworld
- Wisdom & Mischief Café

Modern/Simple:

- The Cat House by Lisa
- L's Cats & Coffee
- The Divine Cat Co.

My top three recommendations would be:

1. Mount Olympus Café (clever wordplay + reflects the theme)
2. The Purrropolis (memorable and unique)
3. Hades & Athena's Hideout (personal touch + catchy)

Would you like me to elaborate on any of these options or brainstorm more ideas?

What about the mythology?

 Great idea! Incorporating mythology into the name can add a unique and intriguing touch, especially with Hades and Athena as inspirations. Here are some name suggestions that draw on mythology while still being fun and inviting for a cat café:

1. **Athena's Cat Sanctuary**
2. **Hades' Purrs & Paws**
3. **The Feline Pantheon**
4. **Olympus Cat Café**
5. **Paws of the Gods**
6. **The Catacombs of Hades**
7. **Athena's Whisker Wisdom**
8. **Hades' Feline Lounge**
9. **Mythical Meows Café**
10. **The Cat's Olympus**
11. **Purrs of the Divine**
12. **Lisa's Mythical Cat Café**
13. **The Enchanted Feline Realm**
14. **Hades & Athena's Mythical Café**
15. **The Catnip Chronicles of Olympus**

These names blend the charm of a cat café with the allure of mythology, making them memorable and thematic!

Notes: **Claude** recognized the mythological connection without hints and offered several suggestions **GPT**, on the other hand, focused on puns and name combinations, missing the historical reference until prompted..

3rd prompt

Expected: a simple, clear and direct ad copy focused on logo design services. No complex metaphors or implicit details.

Responses:

 Here's a compelling ad copy for your graphic design business: Headline: "Your Brand Deserves More Than Just a Logo" Main Copy: Transform your company's visual identity into a powerful business asset. At [Company Name], we don't just design logos – we craft memorable brand stories that speak to your audience. Why partner with us? • 100% custom designs - no templates, no shortcuts • Strategic approach that aligns with your business goals • Timeless designs that stay relevant as your company grows • Comprehensive brand guidelines for consistent implementation [Social Proof] "They didn't just deliver a logo; they delivered understanding. Our new brand identity perfectly captures who we are." • [Client Name], CEO of [Company] Ready to elevate your brand? Book your free consultation today. [Contact Information] Alternative Version (shorter): Is your current logo telling your company's true story? We help established businesses transform their brand identity into their strongest marketing asset. No templates. No cookie-cutter designs. Just strategic, custom-crafted logos that command attention and build trust. [Company Name] - Crafting Brand Legacies Since [Year] Schedule your free consultation: [Contact] [Note: Feel free to modify either version to better match your specific tone and offerings!]	 Unlock Your Brand's Potential with Stunning Logo Design! Is your logo making the right impression? At [Your Business Name], we specialize in creating unique and memorable logos that capture the essence of your brand. In today's competitive market, a strong brand identity is essential for standing out. Our team of experienced graphic designers understands the nuances of effective branding and is dedicated to helping established businesses like yours elevate their visual identity. Why Choose Us? • Tailored Designs: We take the time to understand your business, values, and target audience to create a logo that truly represents you. • Professional Quality: Our designs are crafted with precision and creativity, ensuring your logo is not only beautiful but also versatile across all platforms. • Collaborative Process: We believe in working closely with you throughout the design process, incorporating your feedback to achieve the perfect result. • Timely Delivery: We respect your time and are committed to delivering high-quality designs on schedule. Don't let an outdated logo hold your brand back. Let us help you create a logo that resonates with your audience and reflects your business's unique personality. Ready to transform your brand identity? Contact us today for a free consultation and let's bring your vision to life! [Your Contact Information] [Your Website URL] [Social Media Links] Feel free to customize the copy with your business name and specific details to make it your own!
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Notes: both AIs delivered acceptable responses, but **Claude** once again produced a slightly more structured and complete version.

➤ Conclusion

This analysis highlights how prompt clarity and specificity can significantly impact the quality of AI-generated content. **Claude** consistently demonstrated a better understanding of context and nuance, especially when creative or inferential thinking was required. **GPT-4o mini** performed well but often needed further clarification to fully meet expectations. Overall, crafting well-structured prompts and understanding each model's strengths are key to optimizing content generation.

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